SPRINT 5 REVIEW

During sprint 5, we focused on creating the UI and functionality of the project up to a point where it is presentable and can be demonstrated. We accomplished the main goals we set for this sprint. Most of our work revolved around organising the SUBTLEX dataset according to its (zipf-value) into three levels – Beginner, Intermediate, and Advanced. After classifying the words, we displayed these levels on the library page, where users can preview a word, its definition, an example sentence, and its synonyms. We also made an activity center, which is where users can test what they have learned. For this sprint, we completed the quiz feature, allowing users to answer questions and receive results based on how well they remember the words they studied. This gives users feedback on learning and helps them track their vocabulary progress.

We continued improving general navigation throughout the app. Users are now able to login and immediately see their library and move easily between Library and Activities as well as their progress on the header of Library. First-time and returning users can view a preview of six words. We chose six words because research shows that people can effectively learn about six to seven new vocabulary words at a time. We selected the top six words in each level to help users learn at a manageable and successful pace. Overall, we successfully worked on UI improvements, structured the levels, connected Firestore efficiently, completed the quiz feature, and enhanced the user experience.

For challenges and roadblocks, there was one big issue with Firebase. Specifically, how the free tier limits the amount of reads and writes that are able to be done in a day. We had an issue where our code was doing many reads at once and because of it the entire project stopped working. We basically lost an entire day due to a couple mistakes because when the quota is hit for one of the reads or writes, Firebase just locks everything and makes it so that anything that uses the database just breaks. The next day we get our daily amount of reads and writes but not being able to test as much as we want is an obstacle to get over. We are currently still looking for a solution that would make it so that it doesn't do that but it's very difficult because there are actual consequences for making small mistakes which could potentially kill all the progress we can do for that current day.

Another roadblock was the difficulty in dividing tasks evenly among team members. Some features required more time and effort from certain individuals, which made it challenging to keep the workload balanced. Though this is expected in group work, it still had an impact on our overall speed.

Another roadblock is time and organisation. Because of Firebase limitations and uneven work distribution, some tasks took longer than planned. Because of this, we reorganized and prioritized certain parts of the sprint.

At this stage, we do not yet have official or significant client feedback for Sprint 5. We plan to meet with our stakeholder early in the next sprint to receive updated comments and guidance. We were trying to get the project into a state where it is presentable so that the client/stakeholder will be able to actually have a model and can think of what he expects based on a physical model rather than just a couple of spoken ideas. We are continuously working on the feedback he gave us last time and we are updating Clickup as we go, because many things were removed and there are still things to do on the backlog. Going into the next sprint the objective is to finish what is left of the backlog because it's the last sprint or at least finish as much as we can. Our remaining backlog items include completing the Activity Center by adding matching games, flashcards, and other interactive study tools; expanding the user progress page so people can track their growth across levels; improving account management features, such as saved words and quiz history; and optimizing Firestore calls even further to stay within limits and improve speed. We will continue building on the foundation we created in Sprint 5 as we prepare for a more complete and polished product experience in Sprint 6.